EXCEL LOOKUP FUNCTIONS

Practical Handbook | Lessons 77–87 | Course Videos 142–152

VLOOKUP · INDEX + MATCH · XLOOKUP

A structured, black-and-white reference covering lookup syntax, examples, outputs, data-layout rules, common errors and practical business applications.

Coverage	Details
Lessons	11 sequential lessons (77 through 87)
Topics	VLOOKUP basics and table arrays; numerical and wildcard lookups; two-way matching; approximate tiers; INDEX and MATCH; XLOOKUP dates, horizontal data, higher/lower thresholds and wildcard searches
How to use	Read each chapter in order. Copy sample records into Excel, then enter the formulas in the stated ranges. Calculations use example data and illustrative business rules.

REFERENCE GUIDE

Unless stated otherwise: Excel formulas use English function names and commas as separators; lookup keys should be uniquely identifying when the result must be unambiguous; use real date values for date lookups. Some newer XLOOKUP formulas require a supported Excel version.

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LESSON 77 | How VLOOKUP Works

Goal: search the first column of a selected table range and return a corresponding value from a column to its right.

Core concepts

• Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, FALSE).

• The column number is counted within the table array, starting at 1.

• FALSE requests an exact match; #N/A appears when no exact match exists.

• Dollar signs lock the table array while copying formulas.

Copyable sample data

Fruit name	Quantity	Vendor	Purchase month
Apple	10	Amazon	January
Kiwi	8	BigBasket	February
Grapes	15	Walmart	March

Worked formulas and outputs

Reference table: put Fruit name / Vendor / Purchase month in H4:J7. Search Apple in A2.

```
=VLOOKUP(A2,$H$5:$J$7,2,FALSE)
```

Result: Amazon | Vendor is column 2 of H:J.

Find purchase month.

```
=VLOOKUP(A2,$H$5:$J$7,3,FALSE)
```

Result: January | Purchase month is column 3.

Search a fruit not in the reference data.

```
=VLOOKUP("Banana",$H$5:$J$7,2,FALSE)
```

Result: #N/A | No exact key is present.

Common pitfalls and checks

• The first selected column must contain the lookup key.

• Fixed column indices can become wrong if columns are inserted or rearranged.

Key takeaway

VLOOKUP is useful for conventional left-to-right table lookups, especially in existing workbooks.

LESSON 78 | Table Array Dynamics and Limitations

Goal: choose table arrays accurately and understand why a conventional VLOOKUP cannot return a value to the left of its lookup column.

Core concepts

• If the reference columns are Vendor | Fruit name | Purchase month, a lookup keyed on Fruit name must start at the Fruit name column.

• Column indices are relative to the first column of the selected range; selecting I:J means I = 1 and J = 2.

• Trying to request column 3 from a two-column range returns #REF!.

Copyable sample data

H: Vendor	I: Fruit name	J: Purchase month
Amazon	Apple	January
BigBasket	Kiwi	February
Walmart	Grapes	March

Worked formulas and outputs

Put this reference in H4:J7, and Apple in A2; H:J incorrectly starts at Vendor.

```
=VLOOKUP(A2,$H$5:$J$7,2,FALSE)
```

Result: #N/A | Search is against Vendor rather than Fruit name.

Select only I:J to retrieve purchase month.

```
=VLOOKUP(A2,$I$5:$J$7,2,FALSE)
```

Result: January | Fruit name is now the first selected column.

Invalid index beyond the selected range.

```
=VLOOKUP(A2,$I$5:$J$7,3,FALSE)
```

Result: #REF! | The range has only two columns.

Common pitfalls and checks

• No valid column index within I:J can retrieve Vendor from H.

• Rearrange the columns or use INDEX + MATCH / XLOOKUP for left-side results.

Key takeaway

The lookup key must occupy the first column of VLOOKUP’s table array, and the return column must be within that range.

LESSON 79 | VLOOKUP with Numeric Lookup Values

Goal: perform exact lookups using numeric prices and understand the importance of data types and unique keys.

Core concepts

• Numbers work as lookup values just like text.

• Prices are often non-unique: VLOOKUP returns the first matching row.

• Numeric values stored as text and tiny floating-point differences can prevent exact matching.

Copyable sample data

H: Price	I: Payment	J: Month
7.85	Cash	January
3.25	Card	February
6.35	UPI	March
2.50	Cash	April
4.80	Card	May

Worked formulas and outputs

Put the price table in H4:J9; with 7.85 in B2, return month.

```
=VLOOKUP(B2,$H$5:$J$9,3,FALSE)
```

Result: January | A numerical exact match.

Return payment method.

```
=VLOOKUP(B2,$H$5:$I$9,2,FALSE)
```

Result: Cash | The return column is column 2.

Look up a price not in the table.

```
=VLOOKUP(1.75,$H$5:$J$9,2,FALSE)
```

Result: #N/A | No matching price.

Common pitfalls and checks

• Use =ISNUMBER(B2) to verify a real number; =VALUE(H5) converts numeric text when appropriate.

• Use transaction or product IDs instead of prices whenever records must be unique.

Key takeaway

The matching rule does not change for numerical lookups; the quality and uniqueness of the lookup key matter.

LESSON 80 | Wildcards within VLOOKUP

Goal: match text prefixes, suffixes and patterns when the full lookup text is unknown.

Core concepts

• The asterisk (*) matches zero or more characters; ? matches one character; ~ escapes a literal wildcard.

• To find text beginning with the size abbreviation in C2, concatenate C2&"*".

• VLOOKUP wildcard matching works with FALSE (exact-match mode) and returns the first match.

Copyable sample data

H: Size name	I: Profit rate	Example selling price
Small	13%	1000
Medium	17%	1500
Large	15%	2000
XLarge	10%	2500

Worked formulas and outputs

Put lookup data in H4:I8; size code S in C2 and price 1000 in D2.

```
=VLOOKUP(C2&"*",$H$5:$I$8,2,FALSE)
```

Result: 13% | S* matches Small.

Calculate profit directly.

```
=VLOOKUP(C2&"*",$H$5:$I$8,2,FALSE)*D2
```

Result: 130 | 13% of 1000.

Other pattern formats.

```
"*Laptop" | "*Pro*" | "A?1"
```

Result: Suffix | Contains | One-character pattern | Use appropriate patterns for text.

Common pitfalls and checks

• Broad patterns can match several rows; the earliest match wins.

• If the size code XL matches XLarge, use XL* instead of X* for specificity.

Key takeaway

Wildcards are useful for partial names but should not replace unique identifiers where exactness matters.

LESSON 81 | Two-Way Lookup with VLOOKUP and MATCH

Goal: retrieve a value using an employee ID for the row and a month heading for the column.

Core concepts

• MATCH with match_type 0 returns the exact position of a heading in a row.

• The output of MATCH can replace the fixed col_index_num in VLOOKUP.

• The MATCH header range must begin at the same column as the VLOOKUP table range.

Copyable sample data

Employee ID	Name	Jan	Feb	Mar	Apr
EMP101	Ravi	12000	15000	18000	20000
EMP102	Priya	20000	22000	25000	28000
EMP103	Arjun	16000	19000	21000	23000

Worked formulas and outputs

Paste employee data in A1:F4; find the Feb heading position.

```
=MATCH("Feb",A1:F1,0)
```

Result: 4 | Feb is the fourth header in the sample.

With EMP102 in H2 and Feb in I2, use:

```
=VLOOKUP(H2,$A$2:$F$4,MATCH(I2,$A$1:$F$1,0),FALSE)
```

Result: 22000 | MATCH returns column 4.

Using the short heading Feb as printed in the table:

```
=VLOOKUP("EMP102",$A$2:$F$4,MATCH("Feb",$A$1:$F$1,0),FALSE)
```

Result: 22000 | Use an exact match to the actual header text.

Common pitfalls and checks

• Mismatch between full and abbreviated month names causes #N/A.

• The lookup key still has to be in the first VLOOKUP column.

Key takeaway

VLOOKUP + MATCH adapts dynamically to the selected return-column heading.

LESSON 82 | Approximate VLOOKUP for Grades and Commissions

Goal: map numeric values to thresholds such as student grades and sales commission tiers.

Core concepts

• TRUE performs approximate VLOOKUP and chooses the largest threshold less than or equal to the lookup value.

• The first column of the threshold table must be sorted in ascending order.

• Below the first threshold, VLOOKUP returns #N/A; include a zero baseline when appropriate.

Copyable sample data

H: Minimum marks	I: Grade	Sales threshold	Commission rate
0	F	0	0%
35	E	5000	3%
60	D	10000	5%
70	C	20000	8%
80	B	30000	10%
90	A		

Worked formulas and outputs

Place grade thresholds in H4:I10. Find grade for 78.

```
=VLOOKUP(78,$H$5:$I$10,2,TRUE)
```

Result: C | Largest lower threshold is 70.

For commission, place sales thresholds in K4:L9 (five records).

```
=VLOOKUP(8000,$K$5:$L$9,2,TRUE)
```

Result: 3% | Largest lower threshold is 5000.

Compute commission on 8000 of sales.

```
=8000*VLOOKUP(8000,$K$5:$L$9,2,TRUE)
```

Result: 240 | 8000 multiplied by 3%.

Common pitfalls and checks

• Sort entire reference tables, never only the threshold column.

• Do not confuse lower-threshold matching with choosing the numerically closest number.

Key takeaway

Approximate VLOOKUP is suited to sorted minimum-threshold business rules.

LESSON 83 | INDEX and MATCH

Goal: find a record position using MATCH and return a corresponding field using INDEX, including values to the left.

Core concepts

• INDEX(array,row_num,[column_num]) returns the value at the requested relative position.

• MATCH(value,lookup_array,0) returns the position of the first exact match.

• Lookup and return ranges must have matching record order and boundaries.

• Two MATCH functions can make the row and column dynamic.

Copyable sample data

Employee ID	Name	Region	Net sales
EMP001	Ravi	South	57000
EMP002	Priya	North	43000
EMP003	Arjun	West	75000
EMP004	Sneha	East	96000
EMP005	Amit	South	50000

Worked formulas and outputs

Place sample in A1:D6. Return the fifth employee name.

```
=INDEX(B2:B6,5)
```

Result: Amit | INDEX directly selects row 5 of its array.

Find EMP005 and return employee name.

```
=INDEX(B2:B6,MATCH("EMP005",A2:A6,0))
```

Result: Amit | MATCH provides row position.

Left lookup: search Priya in B, return ID from A.

```
=INDEX(A2:A6,MATCH("Priya",B2:B6,0))
```

Result: EMP002 | Return range may lie to the left.

Two-way: net sales for EMP005.

```
=INDEX(A2:D6,MATCH("EMP005",A2:A6,0),MATCH("Net sales",A1:D1,0))
```

Result: 50000 | Row match = 5, column match = 4.

Common pitfalls and checks

• Header text must match what is actually in the sheet.

• Duplicate identifiers lead MATCH to return the first exact match.

Key takeaway

INDEX + MATCH is a robust and flexible technique that separates the lookup range from the return range.

LESSON 84 | Introduction to XLOOKUP

Goal: replace common VLOOKUP and INDEX + MATCH patterns with shorter, more flexible formulas.

Core concepts

• Syntax: XLOOKUP(lookup_value,lookup_array,return_array,[if_not_found],[match_mode],[search_mode]).

• Exact match is the default; the lookup and return arrays are independent.

• XLOOKUP can look left or right and can return multiple adjacent columns in newer Excel.

• Availability depends on Excel version (Microsoft 365, Excel 2021 or newer supported releases).

Copyable sample data

A: Employee ID	B: Name	C: Department	D: Salary
EMP101	Ravi	IT	65000
EMP102	Priya	HR	55000
EMP103	Arjun	Finance	72000
EMP104	Sneha	IT	68000

Worked formulas and outputs

Place sample in A1:D5. Find EMP103 name.

```
=XLOOKUP("EMP103",A2:A5,B2:B5)
```

Result: Arjun | Default exact matching.

Look left: from Priya to employee ID.

```
=XLOOKUP("Priya",B2:B5,A2:A5)
```

Result: EMP102 | Return array can be to the left.

Not-found handling.

```
=XLOOKUP("EMP999",A2:A5,B2:B5,"Not Found")
```

Result: Not Found | Built-in fourth argument.

Return name, department and salary in one spill range.

```
=XLOOKUP("EMP103",A2:A5,B2:D5,"Not Found")
```

Result: Arjun | Finance | 72000 | Ensure destination cells are empty.

Common pitfalls and checks

• #SPILL! can occur if cells are blocking a multi-column result.

• Check #NAME? if Excel does not support XLOOKUP.

Key takeaway

XLOOKUP is generally simpler for new workbooks and makes left lookups and missing-value handling straightforward.

LESSON 85 | XLOOKUP with Dates and Horizontal Data

Goal: search on true Excel date values and use XLOOKUP across a row, replacing many HLOOKUP formulas.

Core concepts

• Excel dates are numeric serial values; formatted text dates and date-times can differ from date-only values.

• XLOOKUP supports both vertical and horizontal lookup arrays.

• Dates with hidden time portions may require INT to normalize for date-only matching.

• Repeated dates return the first match by default; FILTER is preferable when all matching transactions are needed.

Copyable sample data

A: Date	B: Product	C: Sales	D: Payment
01-Nov-2026	Laptop	45000	Card
03-Nov-2026	Mouse	1200	UPI
05-Nov-2026	Keyboard	2500	Cash
08-Nov-2026	Monitor	15000	Card

Worked formulas and outputs

Create real Excel dates using =DATE(2026,11,5) in F2 and column A.

```
=XLOOKUP(F2,A2:A5,B2:B5,"Not Found")
```

Result: Keyboard | Exact match on date serial number.

Look up sales for same date.

```
=XLOOKUP(F2,A2:A5,C2:C5,"Not Found")
```

Result: 2500 | Return aligned sales value.

Horizontal month row: B1:G1 = Jan through Jun; B2:G2 = 12000,15000,18000,20000,25000,28000.

```
=XLOOKUP("Mar",B1:G1,B2:G2)
```

Result: 18000 | Lookup and return arrays are horizontal.

Date-time column with a time component.

```
=XLOOKUP(F2,INT(A2:A5),B2:B5,"Not Found")
```

Result: Keyboard | INT strips time if dates were stored as timestamps.

Common pitfalls and checks

• Use date values, not merely displayed date-like text.

• For multiple transactions on one day, XLOOKUP returns a single record, not a list.

Key takeaway

One XLOOKUP syntax covers vertical, horizontal and date-based searching.

LESSON 86 | XLOOKUP Higher and Lower Match Modes

Goal: use approximate XLOOKUP to select the proper threshold based on business policy.

Core concepts

• Match mode 0 = exact, -1 = exact or next smaller, +1 = exact or next larger.

• Next smaller picks the largest threshold <= lookup value; next larger picks the smallest threshold >= it.

• Interpret thresholds as minimum qualifications for -1 or upper limits for +1.

• With ordinary linear search mode, XLOOKUP approximate match does not require sorted data.

Copyable sample data

H: Sales threshold	I: Rate
10000	5%
20000	10%
30000	15%
40000	20%
50000	25%

Worked formulas and outputs

Place data in H4:I9. Exact search on 18000.

```
=XLOOKUP(18000,H5:H9,I5:I9,"Not Found",0)
```

Result: Not Found | 18000 is absent.

Next lower qualifying threshold for 18000.

```
=XLOOKUP(18000,H5:H9,I5:I9,"Not Found",-1)
```

Result: 5% | Largest threshold <= 18000 is 10000.

Next higher threshold for 18000.

```
=XLOOKUP(18000,H5:H9,I5:I9,"Not Found",1)
```

Result: 10% | Smallest threshold >= 18000 is 20000.

Calculate amount of next-higher discount.

```
=18000*XLOOKUP(18000,H5:H9,I5:I9,0,1)
```

Result: 1800 | Apply the returned 10% rate.

Common pitfalls and checks

• The correct match mode depends on the definition of the business tiers.

• Binary search modes require data sorted in the correct direction; do not confuse them with match_mode.

Key takeaway

XLOOKUP supports both minimum and maximum threshold policies without needing a helper column.

LESSON 87 | Wildcard XLOOKUP

Goal: perform prefix, suffix and contains lookups in XLOOKUP and choose first versus last match.

Core concepts

• Use match_mode 2 for wildcard matching in XLOOKUP.

• * matches zero or more characters, ? matches exactly one, and ~ escapes wildcard symbols.

• Use G2&"*" for starts-with, "*"&G2 for ends-with, and "*"&G2&"*" for contains.

• With multiple matches, search_mode -1 returns the last matching record.

Copyable sample data

B: Product name	C: Category	D: Price
Samsung Galaxy S24	Smartphone	65000
Apple iPhone 15 Pro	Smartphone	120000
Dell Inspiron Laptop	Laptop	58000
HP Pavilion Laptop	Laptop	62000
Samsung Smart TV	TV	45000
Apple MacBook Pro	Laptop	180000

Worked formulas and outputs

Place product names in B2:B7 and prices in D2:D7. Return first Samsung price.

```
=XLOOKUP("Samsung*",B2:B7,D2:D7,"Not Found",2)
```

Result: 65000 | Match names beginning with Samsung.

Find first product name ending in Laptop.

```
=XLOOKUP("*Laptop",B2:B7,D2:D7,"Not Found",2)
```

Result: 58000 | First matching row is Dell.

Find first product containing Pro.

```
=XLOOKUP("*Pro*",B2:B7,D2:D7,"Not Found",2)
```

Result: 120000 | First matching row is iPhone.

Find last Samsung product.

```
=XLOOKUP("Samsung*",B2:B7,D2:D7,"Not Found",2,-1)
```

Result: 45000 | Reverse search order.

Dynamic search when G2 contains Samsung.

```
=XLOOKUP(G2&"*",B2:B7,D2:D7,"Not Found",2)
```

Result: 65000 | Concatenate the cell value and asterisk.

Common pitfalls and checks

• Wildcards are text-pattern tools; match_mode 2 must be supplied explicitly.

• XLOOKUP returns one matching record; FILTER can return every match.

• Use precise wildcard patterns and unique keys whenever multiple partial matches would be ambiguous.

Key takeaway

XLOOKUP wildcard mode offers flexible partial matching with first- or last-record control.

LOOKUP FUNCTIONS AT A GLANCE

Requirement	Recommended formula	Why
Key in first column, return right	=VLOOKUP(key,table,column,FALSE)	Standard legacy workbooks
Choose a column by its heading	=VLOOKUP(key,table,MATCH(header,headers,0),FALSE)	Dynamic VLOOKUP column
Return from left column	=INDEX(return_range,MATCH(key,lookup_range,0))	Independent return and search ranges
Modern exact lookup	=XLOOKUP(key,lookup_range,return_range,"Not Found")	Default exact match
Numeric minimum threshold	=XLOOKUP(value,thresholds,rates,0,-1)	Choose nearest qualifying lower threshold
Numeric upper-limit threshold	=XLOOKUP(value,thresholds,rates,0,1)	Choose smallest threshold at least as large
Text pattern search	=XLOOKUP("*"&part&"*",names,results,"Not Found",2)	Substring matching
Search the final matching row	=XLOOKUP(key,keys,results,"Not Found",0,-1)	Reverse search

Common errors

Error	Likely reason	First check
#N/A	Lookup not found; data type mismatch; wrong key column	Verify the key, selected first column and exact versus approximate mode
#REF!	Requested index outside selected array	Confirm VLOOKUP column index or INDEX bounds
#VALUE!	Incompatible lookup/return dimensions	Check array lengths and shape
#SPILL!	Blocked multi-cell dynamic array results	Clear destination cells
#NAME?	Function not recognized or misspelled	Check Excel version and spelling

End of Handbook

Lessons 77–87 completed. This handbook is a practical synthesis of the lessons covered in our chat. Data examples are illustrative, and calculations should be checked against your own business definitions before use.